

The schematic shows the SG3524 (U1) configured as a pulse-width modulator. The REF pin (16) is connected to a voltage divider (R8, R9) and a capacitor (C11). The VCC pin (15) is connected to the +12V supply. The IN+ pin (2) is connected to the output of the error amplifier (R10, R12, C14). The IN- pin (1) is connected to the output of the feedback network (R11, R13, C15). The COMP pin (9) is connected to the output of the internal compensator (R13, C15). The RT pin (6) is connected to ground. The CT pin (7) is connected to ground. The OSC pin (3) is connected to the +12V supply. The SD pin (10) is connected to the +12V supply. The CL+ pin (4) is connected to the +12V supply. The CL- pin (5) is connected to ground. The PWR_FLAG pin (12) is connected to the +12V supply. The C_A pin (13) is connected to the +12V supply. The C_B pin (14) is connected to the +12V supply. The E_A pin (11) is connected to the +12V supply. The E_B pin (14) is connected to the +12V supply. The OVP_SD pin (3) is connected to the +12V supply. The ISENSE pin (4) is connected to the +12V supply.

The diagram shows two horizontal green lines representing signal traces. The left trace starts at a red vertical line labeled '+12V' and ends at a red diamond symbol labeled 'PWR_FLAG'. The right trace starts at a red triangle symbol labeled 'GND' and ends at a red diamond symbol labeled 'PWR_FLAG'.

Layout: Q1 / D1 / C4-C6 / R1-R2 loop as small as possible. U1 GND and CL- go to the shunt's ground end (star point), never through the power return. Q1 and D1 on separate heatsinks, or one heatsink with mica + bushings (both tabs are live: drain = SW, cathode = VOUT).

Single-sided board, CNC isolation-milled with a 0.8 mm end mill. THT only, all pad gaps >= 0.8 mm. All electrolytics 50 V. Electrolytic, 5 W resistor and toroid footprints are provisional: measure the parts before placing.

Sheet: / File: monitor-boost.kicad_sch		
Title: Monitor supply – 12 V to 19 V / 3 A boost (SG3524)		
Size: A3	Date:	Rev:
KiCad E.D.A. 10.0.6		Id: 1/1